3.1.1 (b) 27 is not a perfect Square number, Since there doesn't exit an integer whose is 27 so 27 $\notin$ B.

3.1.2 bdh

(b) $\{15\} \subset A$ (true)

$\{15\}$ is a Singleton Set Continous only $15 \in A$

(d) $A \subseteq A$ (true)
Since every set is a subset of itself therefore $A \subseteq A$ is true

(b) $|E| = 3$ (true)
E has 3 elements thus $|E| = 3$.

3.1.3 b

(b) True

3.1.4 F

(F) False

∵ P(C) is set of all subsets of C

∴ $\{4\} \in P(C)$ as $\{4\} \subseteq C = C$

but 4 elements and not a subset is C

∴ $4 \notin p(c)$.

3.1.3 (b)

(b) the cardinality is 6

3.2.1 bdF

(b) $\{2\} \subseteq X \rightarrow$ True

(d) $3 \in X \rightarrow$ False

(F) $\{1,2\} \in X \rightarrow$ True

3.2.2 b

(b) The power set in roster form is

$P(\{1,2\}) = \{\{\}, \{1\}, \{2\}, \{1,2\}\}$.

3.3.1 bd

(b) $\{1,4\}$

(d) $\{-3,0,17\}$

3.3.4 b

(b) P (AUB)

(AUB) means element in A or B

P (AUB) = $\{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}\}$

3.4.3 bdh

(b) $A \cap B = \{3,9\}$ $A \cap C = \{3\}$

$|A \cap B| = 2$ $|A \cap C| = 1$

the statement is False

(d) $C - B = \{2,4\}$

2 and 4 are not multiple of A. So they don't belong in Set A.

The statement is False

(h) $\{3\} \in P(e)$

For any set S, P(S) denotes power set of S
it is set all subsets of S.
the statement is true

3.S.1 bd

(b) Absorption laws : $A \cup (A \cap B) = A$

Hence, complementing on both sides also satisfied as $A' = A'$

(d) Involution or double Complement law

$(A^c)^c = A$

3.S.2 d

(d) $\bar{A} \cap (A \cup B)$

$= (\bar{A} \cap A) \cup (\bar{A} \cap B)$ Distributive law

$= Q \cup (\bar{A} \cap B)$ Complement law

$= \bar{A} \cap B$ (proved) Identity law

3.5.4 b

(b) $(A-B)-A = \emptyset$

$(A-B)-A = (A-B) \cap A'$	Substraction law
$(A-B) \cap A' = (A \cap B') \cap A'$	Substraction Law
$(A-B) \cap A' = (A \cap A') \cap B'$	Associative Law
$(A \cap A') \cap B' = \emptyset \cap B'$	Compliment law
$\emptyset \cap B' = \emptyset$	Compliment law

Hence proved.

3.6.1 (b)

(b) $\{\{foam, tail, non\text{-}fat\}\}$

3.6.4 b

(b) Set $A = \{0, 1\}$

$A^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$

The set A^3 contains all the possible values provided by the set A.

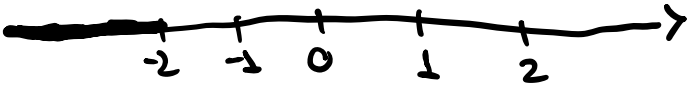
3.6.5 b

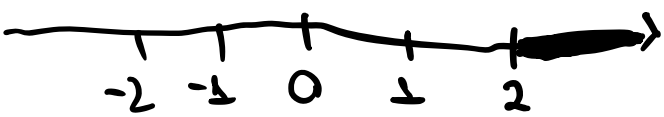
(b) Cordinality of a set defined by a Cartesian

product

$|\{ab\ cd\}\ 3| = |\{\{a\}\ \{b\}\}\{\{c\}\ \{d\}\}| = 4$

Define the set A, B, C, D and E as follows

$A = \{x \in \mathbb{R} / x < -2\}$ 

$B = \{x \in \mathbb{R} / x > 2\}$ 

3.7.1 b

(b) $B = \{2,3,4\}$ $\quad D = \{1,2,3,4,5,6\}$

$C = \{5\}$ $\quad B \cup C = \{2,3,4\} \cup \{5\}$

$B \cup C = \{2,3,4,5\} \neq D$

$\{B, C\}$ does not form petition $\cap \neq \emptyset$ set D.

3.7.3 b

(b) Yes

Since A, B and D are non-empty sets.

$A \cap B = B \cap D = A \cap D = \emptyset$

$A \cup B \cup D = R$, the set of Real numbers

4.1.1 b

(b) Target: $\{w, x, y, z\}$

4.1.2 b

(b) $\{0,1\}^2 = \{00, 01, 10, 11\}$

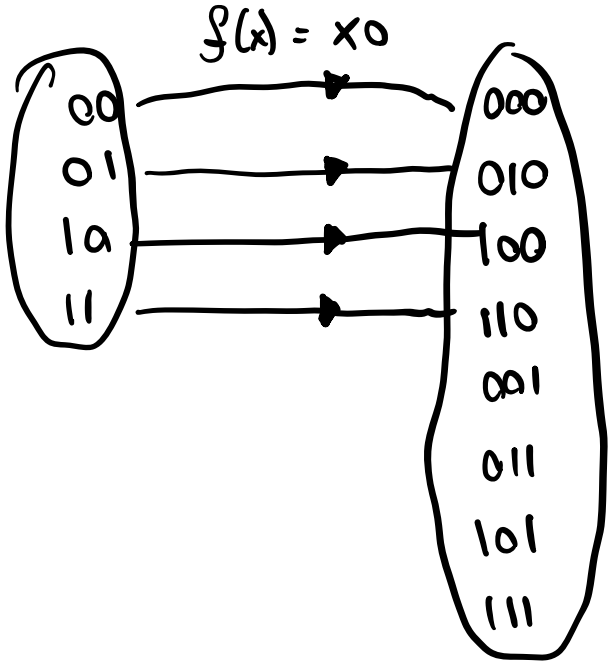
$\{0,1\}^3 = \{000, 001, 010, 100, 011, 111, 110, 101\}$

$f(x) = x0$

$f(00) = 000$, $f(01) = 010$

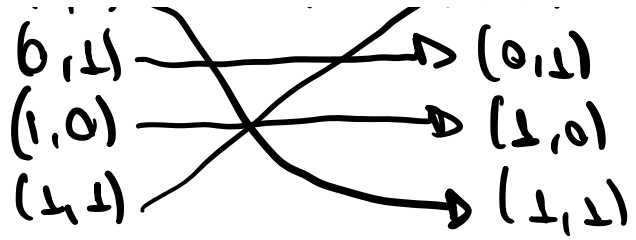
$f(10) = 100$, $f(11) = 110$

Range $f = \{000, 010, 100, 110\}$



4.1.4 b

(b) $[0,0) \xrightarrow{f} (0,0)$



4.1.5 b

(b)

$A = \{2, 3, 4, 5\}$

$f: A \to Z$ such that $f(x) = x^2$

$f(2) = 2 * 2 = 4$

$f(3) = 3 \times 3 = 9$

$f(4) = 4 \times 4 = 16$

$f(5) = 5 \times 5 = 25$

Range $(f) = \{4, 9, 16, 25\}$

4.2.2 b

(b) A backer is packing 8 cookies in one box. He has y cookies to put into boxes.

Number of boxes = $(\frac{y}{8})$

Function will be $f(y) = (\frac{y}{8})$

Where $f(y)$ = number of boxes of cookies.

4.2.3 cd

(c) $\lfloor 5 \rfloor = 5$

(d) $\lfloor \lfloor 3-5 \rfloor - 4.3 \rfloor = \lfloor 3-4.3 \rfloor$

$= \lfloor -1.3 \rfloor$

$= -2$

4.3.1 b

(b) $f(x,y) = |x| - |y|$

let $z \in \mathbb{Z}$

if $z = 0$ then $f(0,0) = 0$

if $z > 0$ then $f(z,0) = z$ ($\because f(z,0) = |z| - 0$)

if $z > 0$ then $f(0,z) = z$ $= 2 - 0$

So, f is onto

4.3.2 bF

(b) $g: \mathbb{R} \to \mathbb{R}$, $g(x) = x^3$

if $g(a) = g(b)$, $a^3 = b^3 \to a = b$. F is one-to-one

We can always find $g(x) = x^3$ for all x. F is onto

therefore, F is both one-to-one and onto

(f) $f: \mathbb{Z} \to \mathbb{Z}$

$f(x) = x - 4$

① let $x_1, x_2 \in \mathbb{Z}$ such that $x_1 \neq x_2$

$\to x_1 - 4 \neq x_2 - 4$

$\rightarrow f(x_1) \neq f(x_2)$

So, $x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$

thus f is one-one

(ii) let $y \in \mathbb{Z}$

now, $x - 4 = y \rightarrow x = y + 4$

$y + 4 \in \mathbb{Z}$ for every $y \in \mathbb{Z}$

So, for every $y \in \mathbb{Z}$ $f(y+4) = y$

thus for every element of $\mathbb{Z}$ there is a pre image under f

Hence f is onto.

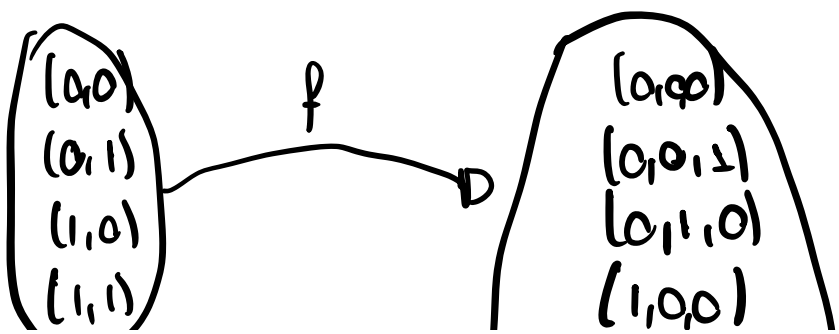
4.3.3 b

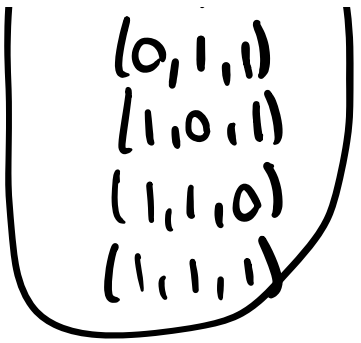
(b) Given the function

$f: \{0,1\}^2 \rightarrow \{0,1\}^3$

Domain of f is $\{(0,0), (0,1), (1,0), (1,1)\}$

Target of f is $\{(0,1,1), (1,0,1), (1,1,0), (1,1,1)\}$





4.4.1 b

(b) Here F is a bijective function so the inverse is well defined, and it is.

$v \to b$

$w \to d$

$x \to e$

$y \to a$

$z \to c$

4.4.2 b

(b) As for $x, y \in \mathbb{Z}$, $f(x) = f(y) \to 2x + 3 = 2y + 3$

Canceling 3 from both sides and dividing by 2 we get $x = y$, that is f is one-one.

When $f(x) = 2x + 3 = y$, then $x = \frac{y-3}{2}$,

But for some $y \in \mathbb{Z}$, $\frac{y-3}{2} \notin \mathbb{Z}$, as when $y = 2$

$\frac{y-3}{2} = \frac{2-3}{2} = -\frac{1}{2} \notin \mathbb{Z}$

and Hence inverse function is not well defined

4.S.1 bd

(b) $\{1,2,3,4,5\}$

(d) Range is $g(f(x))$ in S i.e. $\{1,3,4\}$

4.S.2 bd

(b) $f \circ h(52) = f(h(52)) = f(\lceil \frac{52}{5} \rceil) = f(11) = 11^2 = 121$

(d) $h \circ f = h(f(x)) = h(x^2) = \lceil \frac{x^2}{5} \rceil$

4.S.3 b

(b) Not defined.

4.S.5 b.d

(b) domain of $h \circ g = \{a, b, c\}$

(d) domain of $h^{-1} \circ h = \{1, 2, 3, 4\}$

4.S.8 bd

(b) qol

$g(f(x)) = g(2x+3)$

$5(2x+3)+7$

$10x+15+7$

$10x+22$

$gof = 10x+22$

(d) hof

$h(f(x))$

$h(2x+3)$

$(2x+3)^2+1$

$4x^2+9+12+1$

$hof = 4x^2+12x+10$

5.1.1 b

(b) Because its has the same number of rows and Columns.

5.1.2 b

(b) $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = [a_{ij}]$

$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix}$

$$\begin{bmatrix} 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

So A can be taken as $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

5.1.5 b

(b) 3x4

5.2.1 b

(b) True

5.2.2 b

(b)

5.3.1 b

(b) FALSE

5.3.2 bd

(b) Undefined

(d)

5.3.5 b

(b)